

SMART WIRELESS TANK SENSOR

Complete User & Installation Manual (उपयोगकर्ता और इंस्टालेशन गाइड - Hinglish)

MODEL: CH32V003-LORA V2.0

WIRELESS RANGE

433 MHz LoRa (~1.5 km)

BATTERY LIFE

Up to 1 Year (Ultra Low Sleep)

PROBE TECHNOLOGY

Zero-Corrosion Pulsing

SUPPORTED TANKS

Tank 1, Tank 2, Tank 3

1. Yeh Sensor Kya Hai aur Kaise Kaam Karta Hai?

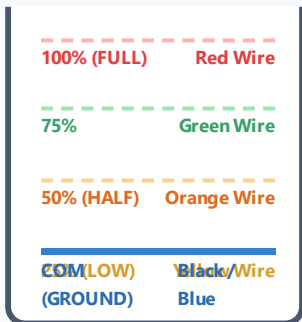
Yeh ek **Smart Wireless Tank Level Sensor** hai jo aapki chhat (rooftop) par lagi paani ki tanki ke andar fit hota hai. Yeh sensor tanki ke paani ka level (0%, 25%, 50%, 75%, 100%) aur apni battery ka status real-time naapta hai aur bina kisi taar (wire) ke niche room me lage **Water Level Indicator Display Unit** ko radio signal (LoRa / 433 MHz) ke zariye bhejta hai.

⚡ Zero-Corrosion (जंग से 100% सुरक्षा) Technology:

Purane sensors me taaron par lagataar current rehta tha, jisse taar 2-3 mahine me gal jaate the aur paani me chemical reaction hota tha. Is naye sensor me P-MOSFET Switch laga hai jo sirf **10 millisecond ke liye micro-pulse** bhejta hai aur foran off ho jata hai. Isse sensor ke probes saalon-saal naye rehte hain aur paani 100% safe rehta hai!

2. Wires & Tank Installation (वायरिंग और लगाने का तरीका)

Sensor unit se total **5 Wires (या Probes)** tanki ke andar jaati hain. Inhe niche diye gaye height ke hisaab se tanki me latkana (suspend) karein:



💡 Installation Tips:

- COM Probe** tanki ke bilkul tale (bottom) par hona chahiye taaki paani hamesha ise touch kare.
- 100% Probe** overflow pipe se 2-3 inch niche rakhein taaki motor overflow hone se pehle band ho sake.
- Probes ko aapas me touch hone se bachayein (aap plastic pipe ya cable ties use kar sakte hain).

Wire Name	Color Code	Tank Position	Kaam / Description
COM (Common)	BLACK / BLUE	Bottom (0%)	Main Reference Wire. Yeh hamesha paani me doobi rehni chahiye.
Probe 25%	YELLOW	1/4th Height	Paani is wire ko chhoote hi display par 25% dikhega.
Probe 50%	ORANGE	Half (1/2 Height)	Aadhi tanki bharnे ka signal deta hai (50%).
Probe 75%	GREEN	3/4th Height	Paani 75% pahuchne par signal deta hai.
Probe 100%	RED	Top (Full)	Tank Full Level. Isko chhoote hi Controller motor ko band kar deta hai.

🔧 Broken Wire Auto-Recovery (तार टूटने पर भी सिस्टम फेल नहीं होगा):

Agar kisi wajah se beech ki koi wire (jaise 50%) toot jaaye ya uspar kachra jam jaaye, lekin 75% par paani pahuch gaya ho, toh sensor ka intelligent algorithm automatically samajh jaata hai ki 50% bhi paani me hai! Display par galat reading aane ke bajaye accurate 75% hi aayega aur sath hi service alert generate karega.

3. Multi-Function Button Guide (बटन का सही इस्तेमाल)

Sensor board par ek hi button diya gaya hai jisse aap 3 alag-alag kaam kar sakte hain. Button dabane ki timing niche dekhein:

CLICK (1 Sec) Force Send / Test		HOLD 3-5 Sec Device Restart	HOLD >5 Sec Factory Reset & Pairing
Button Action	Kitni Der Dabayein?	LED Response	Kya Hoga? (Function)
Instant Force Send (Test)	Sirf 1 baar daba kar chhod dein (1 Second)	LED 1 baar turant chamkegi	Sensor foran paani ka level aur battery naap kar Controller ko bhejega. Testing ke waqt wait nahi karna padta.
Device Reboot (Restart)	3 se 5 second tak dabakar rakhein	LED 2 baar slow blink karegi	Sensor restart ho jayega. (Dhyaan rahe: Isse Pairing ya Tank ID delete NAHI hoti).
Factory Reset & Pairing	5 second se zyada dabaye rakhein	LED lagataar blink karne lagegi	Purani pairing delete ho jayegi aur sensor 1 minute ke liye Pairing Mode me aa jayega.

4. LED Signals ka Matlab (एलईडी संकेत)

LED Status	Matlab (Status)	Aapko Kya Karna Hai?
Ek baar halki chamak (1 Flash)	Data Controller tak successfully bhej diya gaya hai.	Normal condition, kuch karne ki zaroorat nahi.
Do baar dheemi chamak (2 Blinks)	Sensor reboot ho chuka hai aur normal ho gaya hai.	Normal.
Lagataar Blink (Blinking 1 min)	Pairing Mode on hai, Controller se connect hone ka wait kar raha hai.	Controller me jakar "Pair Tank" select karein.
LED Band (OFF)	Sensor Deep Sleep mode me hai (Battery save kar raha hai).	Yeh normal hai. Button dabane ya paani badalne par hi LED on hoti hai.

5. Water Level Indicator ke Saath Pair Kaise Karein? (Pairing Guide)

Naya sensor lagate waqt ya unpair karne ke baad pairing karna bohot aasaan hai:

- Controller Setup:** Niche room me lage Main Display par Menu button dabayein, "**Pairing Settings**" me jayein, aur jis tank me lagana hai use select karein (e.g., "*Tank 1*"). Screen par [SCANNING...] likha aayega.
- Sensor ko Pairing Mode me daalein:** Tank sensor ka button **5 second tak dabaye rakhein** jab tak board ki LED lagatar blink na karne lage.
- Pairing Confirm:** Controller screen par turant [PAIR SUCCESS] aayega, buzzer se 'Beep' sound aayegi, aur tank ka level aur battery % display ho jayega.
- Permanent Memory:** Dono devices apni flash memory me ID permanently save kar lete hain. Bijli jaane par ya battery nikalne par bhi pairing delete nahi hoti!

6. Sensor Kab-Kab Data Bhejta Hai? (Smart Sleep Cycles)

Battery ko saal bhar chalane ke liye sensor me **Adaptive Dynamic Sleep** algorithm diya gaya hai:

- **Normal Time (15s Deep Sleep):** Jab paani ka level ek jaisa rehta hai, sensor 15 second tak gehri neend (Deep Sleep) me rehta hai taaki 0.015mA se kam current kharch ho.
- **Paani Bharte Waqt (5s Fast Track):** Jaise hi motor chalti hai aur paani badalta hai, sensor foran 5-second fast mode me aa jata hai aur agle 1 minute tak har 5 second me display ko real-time update bhejta hai.
- **Anti-Wave Filter (लहरों से सुरक्षा):** Paani girne se paani me jo chhaap/lehrein banti hain, unse galat reading na jaye isliye sensor 2 lagataar cycles me level verify karke hi final data bhejta hai.
- **Heartbeat Telemetry (Har 4 Ghante):** Agar poore din paani ka level bilkul na badle, tab bhi sensor har 4 ghante me Controller ko "*Main Active Hoon*" ka signal aur exact battery percentage report bhejta hai.

7. Battery & Power Advice

Kaisi Battery Lagayein?

Sensor **2.8V se 3.6V DC** par kaam karta hai. Best performance aur lambi life ke liye:

- **Best Option:** 2x 1.5V AA Alkaline Cells (Duracell Ultra / Energizer) jo market me aasani se milte hain.
- **Rechargeable Option:** 1x 3.2V LiFePO4 cell ya 3.7V 18650 Li-ion cell (Ultra-low quiescent LDO ke saath).
- **Low Battery Alert:** Jab battery 20% se kam ho jayegi, Display Unit par battery icon blink karega aur warning dega. Tab cells ko replace kar lein.

? 8. Troubleshooting & FAQs (समस्या निवारण)

Problem	Kyun ho sakta hai?	Solution (Kese Theek Karein)
Display par "DISCONNECTED" ya Signal 0% aa raha hai	Sensor battery khatam ho gayi hai ya antenna theek nahi hai.	1. Sensor button 1 sec click karein aur LED dekhlein. 2. Battery voltage check karein. 3. Sensor ka antenna seedha khada rakhein.
Tanki khali hai par display 25% ya 50% dikha raha hai	Probes aapas me chipak gaye hain ya wire me moisture/kachra hai.	Tanki me taaron ke beech thoda gap banayein aur probes ko kapde se saaf kar dein.
Motor full hone par bhi band nahi hui	100% (Red) wire overflow level se upar lag gayi hai.	100% wire ko overflow pipe ke mouth se 2-3 inch niche adjust karein.
Sensor naye Indicator se connect nahi ho raha	Purana pairing record sensor ke memory me saved hai.	Sensor button ko 5 second se zyada hold karein (LED blink hogi) aur naye display se re-pair karein.